Find out the total cumulative sales after 4 months and the sales during the fifth month. Also find out the total sales due to the advertisement campaign $\left[e^{-1.2} = 0.3012,\ e^{-1.5} = 0.2231\right]$.

Solution:

Assume that $F(t)$ is the total sales after t months, sales rate is $\frac{d}{dt}F(t) = f(t)$

$$\therefore \quad F(t) = \int_0^t f(t)dt$$

Total cumulative sales after 4 months.

$$F(4) = \int_0^4 f(t)dt$$

$$= \int_0^4 3000\, e^{-0.3t}\, dt$$

$$= 3000\left[\frac{e^{-0.3t}}{-0.3}\right]_0^4$$

$$= -10{,}000\ \left[e^{-1.2} - e^{0}\right]$$

$$= -10{,}000\left[0.3012 - 1\right]$$

$$= 6988 \text{ units}$$

(ii) Sales during the 5th month

$$= \int_4^5 3000\, e^{-0.3t}\, dt$$

$$= 3000\left[\frac{e^{-0.3t}}{-0.3}\right]_4^5$$

$$= -10{,}000\left[e^{-1.5} - e^{-1.2}\right]$$

$$= -10{,}000\left[0.2231 - 0.3012\right]$$

$$= 781 \text{ units}$$

Total sales due to the advertisement campaign.

$$= \int_0^\infty 3000\, e^{-0.3t}\, dt = \frac{3000}{-0.3}\left[e^{-0.3t}\right]_0^\infty$$

$$= -10000\left[0 - 1\right]$$

$$= 10{,}000 \text{ untis.}$$

Example 3.15

The price of a machine is 6,40,000 if the rate of cost saving is represented by the function $f(t)$ = 20,000 t. Find out the number of years required to recoup the cost of the function.

Solution:

Saving Cost $S(t) = \int_0^t 20000\, t\, dt$

$= 10000\ t^2$

To recoup the total price,

$10000\ t^2 = 640000$

$t^2 = 64$

$t = 8$

When t = 8 years, one can recoup the price.

3.2.2 Revenue functions from Marginal revenue functions

If R is the total revenue function when the output is x, then marginal revenue $MR = \frac{dR}{dx}$ Integrating with respect to 'x' we get

Revenue Function, $R = \int (MR)\, dx + k.$

Where 'k' is the constant of integration which can be evaluated under given conditions, when x = 0, the total revenue R = 0,

Demand Function, $P = \frac{R}{x}, \quad x \neq 0.$

Example 3.16

For the marginal revenue function $MR = 35 + 7x - 3x^2$, find the revenue function and demand function.

Solution:

Given $MR = 35 + 7x - 3x^2$

$$R = \int (MR)\, dx + k$$

$$= \int \left(35 + 7x - 3x^2\right) dx + k$$

$$R = 35x + \frac{7}{2}x^2 - x^3 + k$$

Since R = 0 when $x = 0, \quad k = 0$

$$R = 35x + \frac{7}{2}x^2 - x^3$$

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